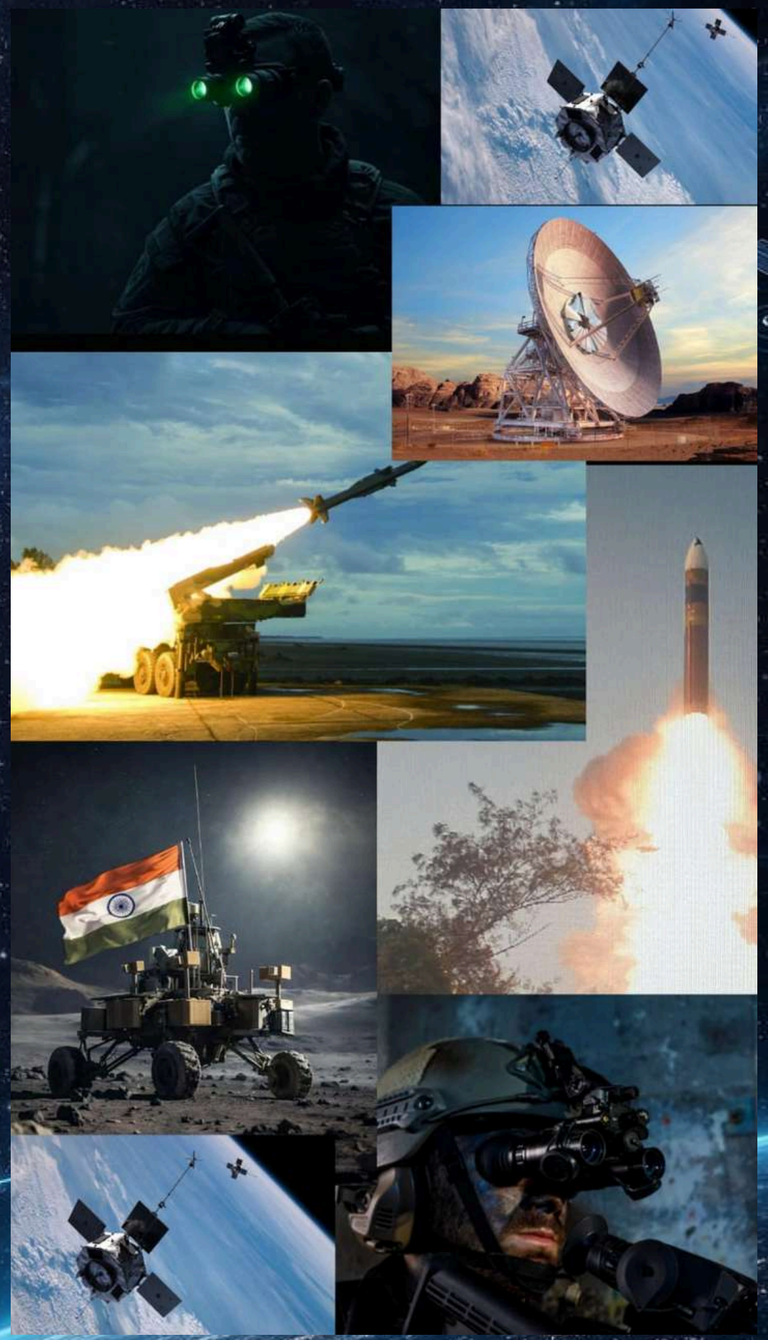
Poster Project Competition

C PROGRAMMING IN SPACE & DEFENSE SYSTEMS



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INTRODUCTION

Our topic focuses on the role of C programming in space and defence systems . C is widely used because it offers high speed, low memory usage, and precise hardware control. In critical missions like spacecraft flight software, missile guidance, radars, and rovers, systems must respond in real time with zero room for failure. This is why C remains one of the most trusted languages for mission-critical applications.

NEED OF C PROGRAMMING IN SPACE

- Limited hardware: Spacecraft don’t have powerful laptops or servers; they use small, rugged computers.
- No second chances: If something goes wrong millions of kilometers away, you can’t fix it.
- Real-time decisions: Navigation, communication, and survival systems must respond instantly.

ROVERS AND LANDERS

Mars rovers like Curiosity and Perseverance use C/C++ for:
- Autonomous driving.
- Operating robotic arms.
- Processing images from cameras.

These systems must make decisions without waiting for Earth commands (since signals take minutes to travel).

NIGHT VISION GOGGLES

C can also be used in army (infantry's) night vision goggles and basically provide them with a grid system (as used in GPS) of unknown houses/places, this will help them in their operations in the dark.
C programming controls night vision goggles by processing sensor signals, enhancing images, reducing noise and enabling real-time target detection. Its speed and reliability support military surveillance and security operations in darkness.

MIRV MISSILE GUIDANCE

- India’s Agni-5 missile with MIRV (Multiple Independently Targetable Re-entry Vehicle) tech uses C for guidance.
- One missile can release multiple warheads at different targets.
- C ensures each warhead gets its own precise path in real time.

SPACECRAFT FLIGHT SOFTWARE

- Every spacecraft has a “brain” called the flight computer.
- C is used to write the software that:
- Controls engines and thrusters.
- Manages power systems (solar panels, batteries).
- Handles communication with Earth.

RADAR SYSTEMS

Radars such as Rajendra Radar (Akash system) and Arudhra Radar rely on C for:

- Signal processing (detecting and tracking targets).
- Synchronizing antenna movement.
- Real-time communication with missile batteries.
- Controls radar hardware (antenna, transmitter, receiver)
- Collects raw echo signals reflected from targets
- Filters noise and unwanted signals using signal-processing logic
- Calculates distance, speed, and direction of targets
- Tracks multiple targets simultaneously in real time
- Sends processed data to display and control systems
- Ensures fast, reliable execution with minimal delay

MISSILE SYSTEMS

- DRDO (Defence Research and Development Organisation) develops missile guidance
 and control systems (e.g., Agni, Prithvi, Akash).
- These systems rely on real-time embedded software, often written in C/C++,
 because

- Guidance algorithms need deterministic execution.
- Hardware integration with sensors, gyroscopes, and actuators requires low-level programming.

UNMANNED SYSTEMS

C enables direct hardware control of motors and sensors in unmanned systems. Its fast, real-time performance supports stabilization and control loops. Widely used in embedded firmware and RTOS applications, C efficiently processes sensor data for navigation and handles reliable communication through UART, I²C, SPI, CAN, and MAVLink protocols.

Rustom UAVs and other drones use C for:

- Flight control software.
- Sensor fusion (combining GPS, IMU, cameras).
- Real-time communication with ground stations.

THANK YOU